EDITORIAL

AI IN MEDICINE

Updated Guidance for Author Use of AI in Medical Publication

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Artificial intelligence (AI) is reshaping research and scientific publishing at an unprecedented pace, with extraordinary promise. We as editors embrace AI as a tool that can be helpful to researchers and authors by reducing the burdens of some tasks and improving efficiency in some contexts. However, as use of AI in scientific publishing grows exponentially,[1-3] so do concerns about such use.[3-10] Journals have adopted policies regarding appropriate and responsible use of AI by authors, reviewers, and editors, which range from encouragement of AI use, provided that such use is disclosed, to restrictions on some specific types of use.[11-13] Yet very low rates of disclosure of AI use by authors have been reported.[14-17] Reasons reported by authors for failure to disclose use of AI include concern that it may lead to biased editorial assessments and unfavorable editorial decisions and lack of understanding of journal policies of what should be disclosed.[18]

Recognizing that transparency is central to the integrity of the scholarly record, recommendations for more uniform disclosure of AI use by authors may be helpful. Efforts to develop guidelines that promote responsible use and disclosure and related initiatives are underway.[19-22] In the interim, we offer guidance to authors for responsible use, highlighting what is permitted and not permitted and what needs to be disclosed (**Table** and **Box**).

In some cases, disclosure policies may not be sufficient to address concerns about AI-generated content that is incorrect or even fabricated (ie, hallucinated). Such policies also cannot stem the tide of AI-generated manuscript submissions with little to no scholarly or clinical value. For example, across the JAMA Network we have received manuscripts with multiple inaccurate or nonexistent references, a common problem when authors use AI to identify or format their references. Even

Table. JAMA Network Journals Guidance for Acceptable Use of AI

Use type	Acceptable use?	Comment
Research	Yes, provided the use is fully described in the Methods section and reporting guidance is followed and authors take full responsibility for all content	See detailed guidance in the Editorial by Flanagin et al[22]
Presenting AI-generated data as if it were original research from non-AI sources	No	
Literature search	Yes, provided the use is fully described in the Methods section for reports of research and the Acknowledgment section for other types of manuscripts and that authors check for accuracy and take full responsibility for all content	
Manuscript preparation	Yes, provided the use is fully described in the Acknowledgment section and authors check for accuracy and take full responsibility for all content	See detailed guidance in the Editorial by Flanagin et al[22] and the Box
Checking grammar or spelling	Yes	No disclosure needed
Translation	Yes, provided the use is fully described in the Acknowledgment section and authors check for accuracy and take full responsibility for all content	Be aware of risk of entering content into public translation solutions
Drafting letters and online Comments	No	Human insights are prioritized
Drafting Opinion manuscripts	No	Human insights are prioritized
Bibliographic references	Discouraged	AI tools may cause problems with the accuracy and integrity of cited references; standard reference managers may be used instead
Formatting data or graphic visualization of research data or results	Yes, provided that use is fully described in the figure legend and authors check for accuracy and take full responsibility for all content	
Clinical images and illustrations	No	
Clinical video and audio	No	
Peer review	No	Entering an author's manuscript or the reviewer's summary of the manuscript into an external AI tool or LLM is a violation of confidentiality
Author response to peer reviewer and editor comments	No	Author's insights are prioritized

Abbreviations: AI, artificial intelligence; LLM, large language model.

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frontier models can introduce realistic-looking references, reflecting real authors and titles that support a particular point but that do not exist. As a result, we have updated our guidance to advise authors to not use AI, including large language models (LLMs), to generate, format, or otherwise manage references. We encourage the use of standard reference managers instead, and we remind authors to check all references for completeness, accuracy, and integrity. The references are an important section of any manuscript, and authors are as responsible for this section as for any other.

Our research and experience also demonstrate that author use of AI to assist with manuscript writing is often associated with specific types of articles, such as Opinion articles and Letters to the Editor.[14] We have seen marked increases in submissions of these types of articles, occasionally from the same authors, who may not have expertise in the topics they are addressing. These submissions are often formulaic and lack useful insights.[23] Thus, we have revised our guidance to authors to inform them that use of AI, including LLMs, to draft Opinion manuscripts, Letters to the Editor, and online Comments is not permitted. Use of AI tools and software to check grammar and spelling or for translation is permitted, but only after the initial text is written by humans. Authors must understand the risks of putting their content into an AI model and take full responsibility for any content changed by such tools.

We also share concerns about the accuracy, clinical fidelity, and demographic representational bias in AI-generated images.[24] Therefore, we have updated our guidance regarding use of AI for clinical images and illustrations to more explicitly state that submission of clinical images and clinical illustrations created or manipulated by AI is not permitted unless part of formal research design or methods, which must be fully disclosed. Preparation of scientific images (clinical images, radiographic images, micrographs, gels, etc) for publication must preserve the integrity of the image data. Digital adjustments of brightness, contrast, or color applied uniformly to an entire image are permissible as long as these adjustments do not selectively highlight, misrepresent, obscure, or eliminate specific clinical or scientific elements in the original image. Replacement of nonclinically relevant background with a solid color in photographs is permitted. Any such adjustment or background replacement must be fully described in the figure legend along with indication of the AI or software (name and version number) used, and the original unedited photographs must be provided for editorial review. Use of AI to generate graphic visualization of research data or results is permitted, provided that the AI or software (name and version number) used is included in the figure legend. The submission of clinical video or audio created or manipulated by AI is not permitted unless part of formal research design or methods, which must be fully disclosed.

Beyond these disclosure requirements and restrictions on AI use for specific types of content, our guidance reaffirms the foundational tenet that authors must be responsible for the accuracy of all content submitted to journals, whether AI generated or assisted or produced solely by humans. We recognize the rapidly evolving and extraordinary influence of AI on research and scientific publishing, with concomitant efficiencies, as well as threats and concerns. When AI is used as part of a scientific study, we ask authors to follow our previously released guidance for complete and transparent reporting of the methods and results to address potential biases, limitations, generalizability, and reproducibility.[22] Our updated guidance for appropriate use of AI in manuscript and other content preparation, and for what is not permitted, is intended to promote and preserve accuracy and integrity of publication at a time when upholding the trustworthiness of the scientific record is more critical than ever. We have included this updated guidance in the *JAMA* and JAMA Network journals' Instructions for Authors,[25] and we will monitor and continue to update this guidance to best support authors and the quality and transparency of publication.

Box. Use of AI in Manuscript and Content Preparation and Submission

Responsibility: Authors are responsible for the accuracy of all content submitted to the journal.

Authors must review and confirm the accuracy of all content generated by artificial intelligence (AI), large language models (LLMs), chatbots, or similar technologies and must be accountable and take full responsibility for all content.

Disclosure: Use of AI, LLMs, chatbots, or similar technologies to assist with content creation, revision, and formatting must be reported in the Acknowledgment section.

References: AI, LLM, and chatbots should not be used to generate or format references because these tools may cause problems with the accuracy and integrity of cited references; standard reference managers may be used instead. Check all references for completeness, accuracy, and integrity.

Opinion and Correspondence Manuscripts: Use of AI, LLM, and chatbots to draft Opinion manuscripts, Letters to the Editor, and online Comments is not permitted.

Clinical Images, Illustrations, and Multimedia: Submission of clinical images and clinical illustrations created or manipulated by AI, LLM, and chatbots is not permitted unless part of formal research design or methods, which must be fully disclosed. This policy also applies to the submission of clinical video and audio files.

Failure to Disclose: If incomplete disclosure or failure to disclose use of AI is detected during the editorial process or after publication, content may be rejected or subject to postpublication corrective action.

If AI, LLM, or chatbots were used to assist with content creation, revision, or formatting, the following should be reported in the Acknowledgment section:

- Name of the AI software platform, program, or tool
- Version and extension numbers
- Manufacturer
- Date(s) of use
- A brief description of how the AI was used and on what portions of the manuscript or content
- Confirmation that the author(s) takes responsibility for the accuracy and integrity of the content generated

Note this guidance does not apply to basic tools for checking grammar or spelling.

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